



ÉCOLE POLYTECHNIQUE DE SOUSSE

DNS Server Deployment & PowerShell-Based Diagnostics

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Chapitre 1

DNS Server Deployment & PowerShell-Based Diagnostics

1.1 Objective

This lab focused on deploying a complete **DNS (Domain Name System) infrastructure on Windows Server**, covering both **forward lookup** (name-to-IP resolution) and **reverse lookup** (IP-to-name resolution), along with essential maintenance mechanisms such as dynamic updates and scavenging of stale records. **PowerShell** was used extensively throughout to validate and diagnose DNS functionality, reinforcing command-line proficiency alongside GUI-based administration.

1.2 1. Lab Environment Preparation

Two virtual machines were configured on the same VMware network (10.10.10.0/24) :

- SRV-DNS1 — Windows Server 2019/2022, static IP : 10.10.10.10
- PC1 — Windows 10/11, static IP : 10.10.10.20

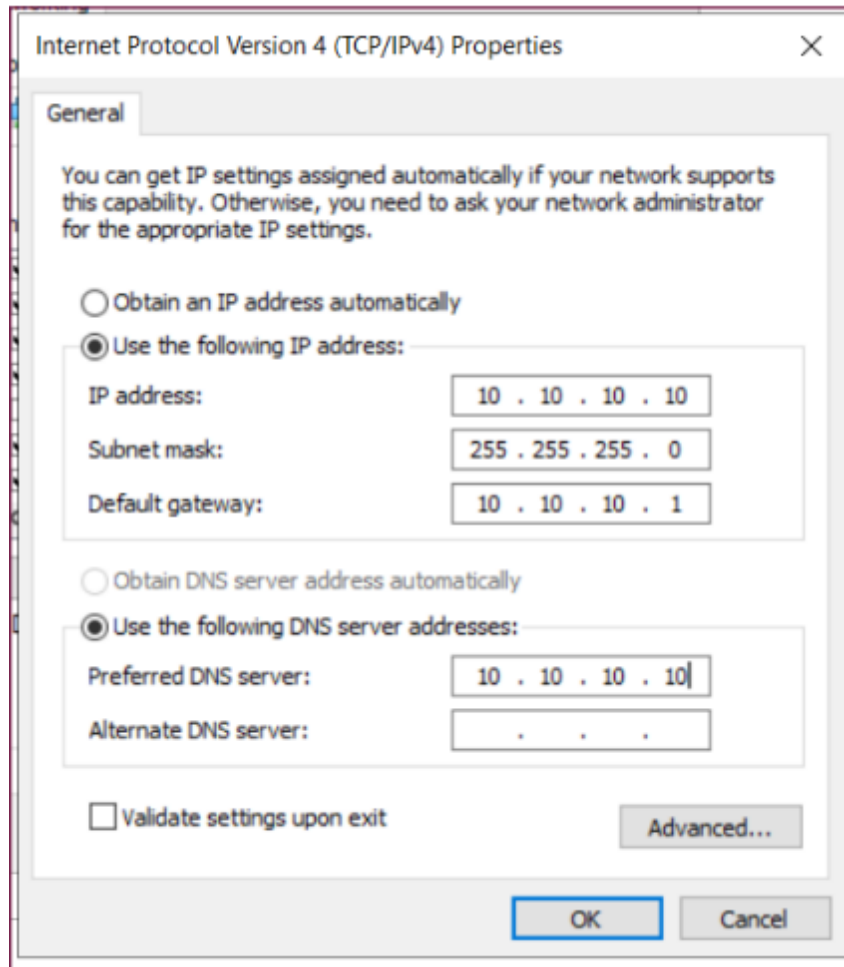


FIGURE 1.1 – Client TCP/IPv4 configuration with static IP and preferred DNS server

1.3 2. Base Server Configuration (GUI)

The server was renamed to SRV-DNS1 via Server Manager → Local Server, a required step before deploying directory-integrated services, followed by a restart to apply the change.

1.4 3. DNS Role Installation

The **DNS Server** role, along with its administration tools, was installed via Add Roles and Features. Installation completed without errors, with the role confirmed alongside the already-installed AD DS and DHCP Server roles — reflecting a consolidated infrastructure services deployment on a single host.

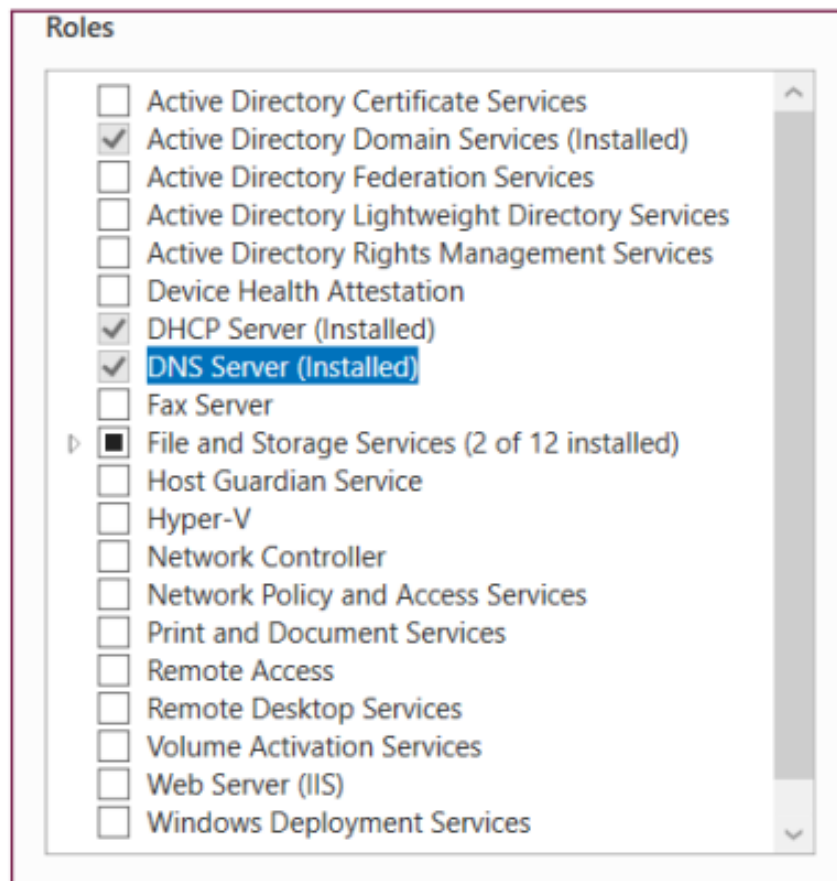


FIGURE 1.2 – DNS Server role installation confirmed alongside AD DS and DHCP Server roles

1.5 4. Forward Lookup Zone Creation

A forward lookup zone was configured with Active Directory-integrated replication, enabling automatic replication of DNS data across domain controllers rather than relying on manual zone transfers — a best practice in AD-integrated environments for both resiliency and security.

A first Host (A) record was added :

- Name : PC1
- IP address : 10.10.10.20
- Option enabled : Create associated PTR record (automatically generates the corresponding reverse-lookup entry, keeping forward and reverse zones in sync)

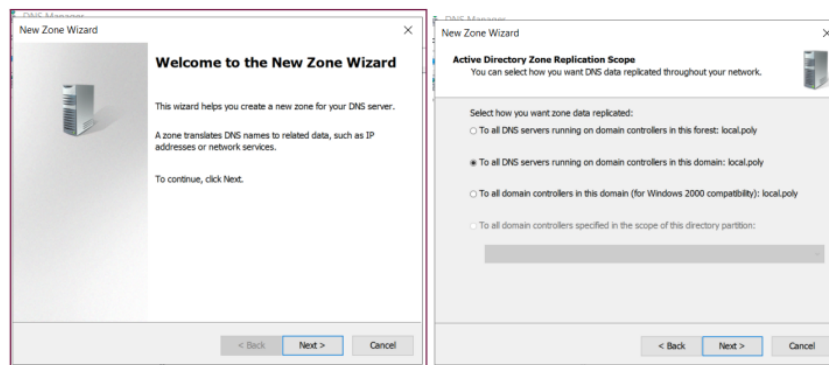


FIGURE 1.3 – New Zone Wizard : launching the forward lookup zone creation and selecting the Active Directory replication scope

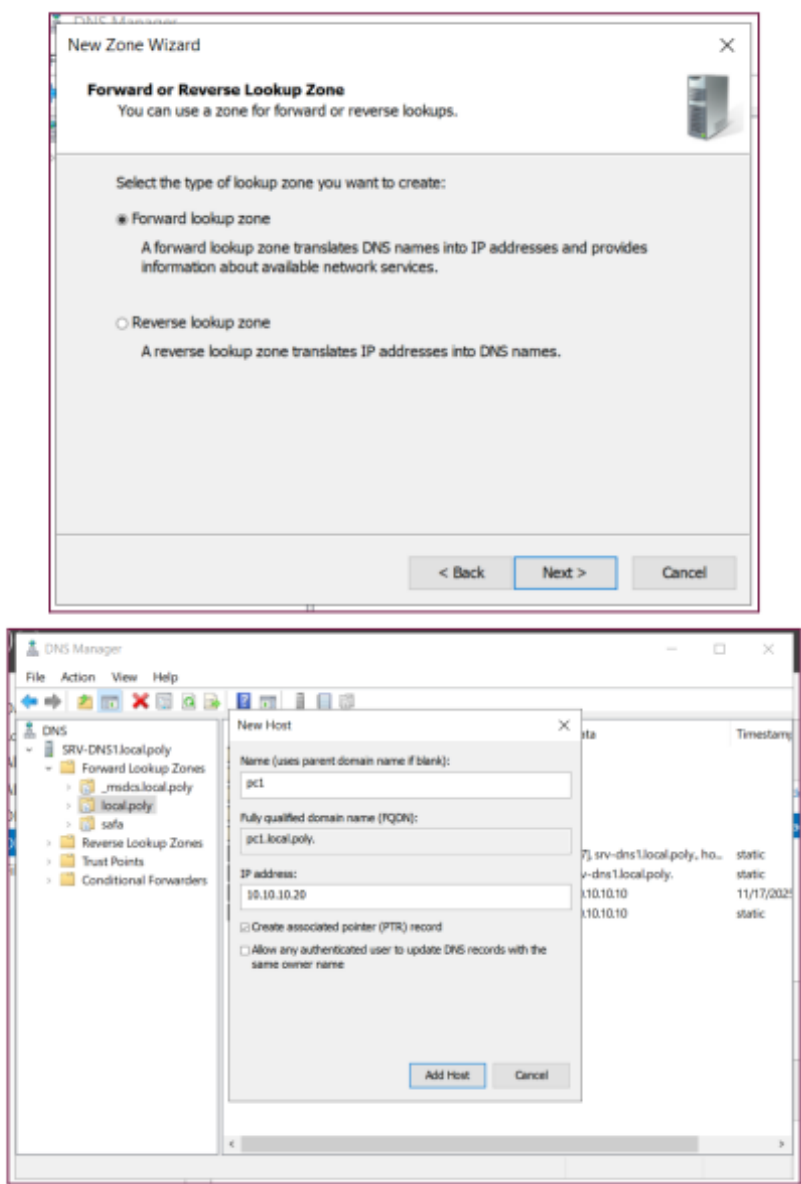


FIGURE 1.4 – Selecting the Forward Lookup Zone type and creating the Host (A) record for PC1

1.6 5. Reverse Lookup Zone Creation

A primary reverse lookup zone was created with :

- Network ID : 10.10.10
- Dynamic update : Secure only

Once created, the PTR record for PC1 was verified to have been generated automatically, confirming the forward-to-reverse record linkage configured in the previous step.

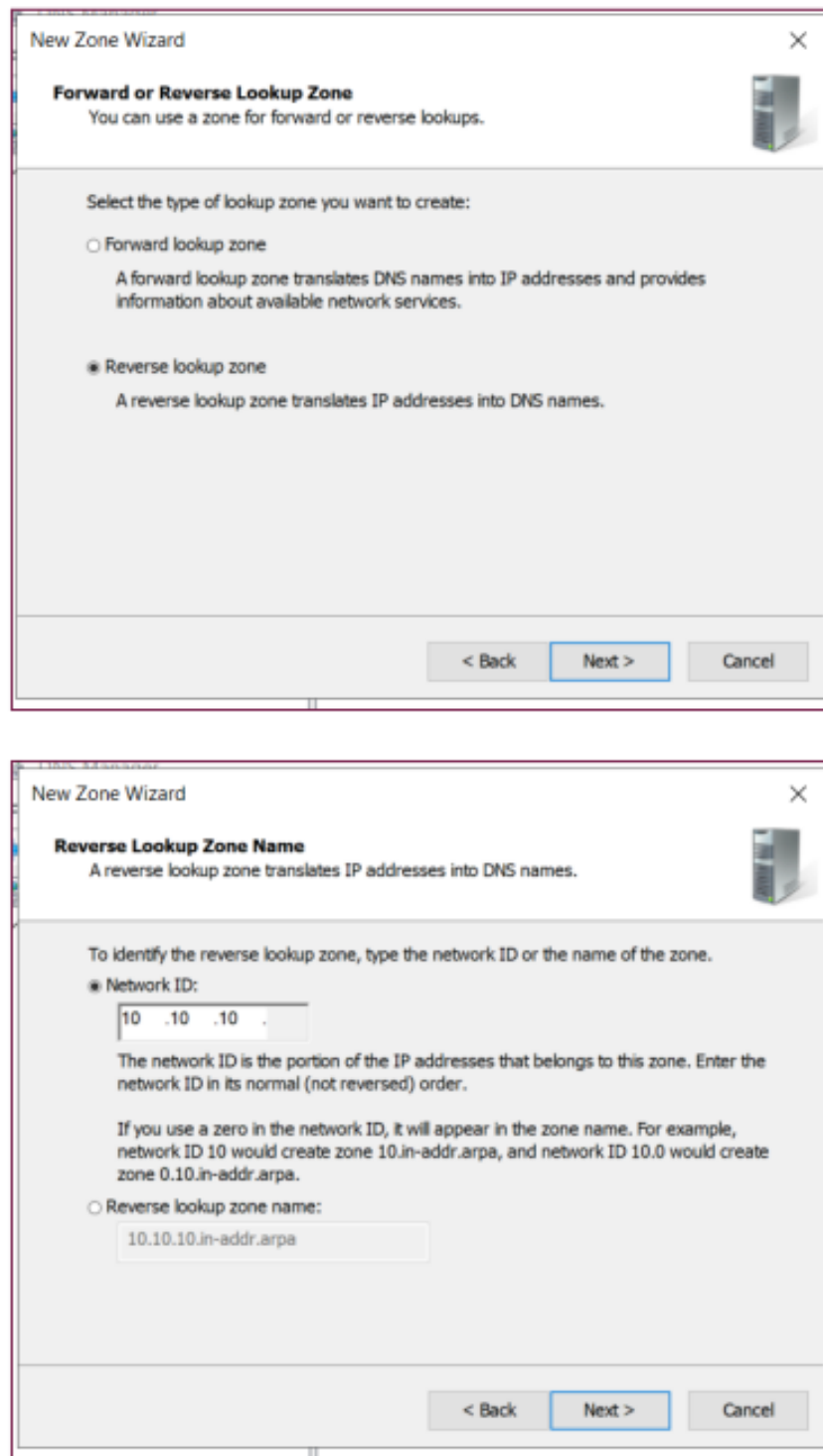


FIGURE 1.5 – Reverse Lookup Zone creation : selecting the zone type and defining the network ID (10.10.10)

1.7 6. Core DNS Resource Records

To simulate a realistic production DNS zone, the following essential record types were created :

- A Record — web → maps a hostname directly to an IP address



FIGURE 1.6 – Confirmation of successful A record creation for web.local.poly

- CNAME Record — alias www pointing to web.poly.local, allowing multiple names to resolve to the same host without duplicating A records

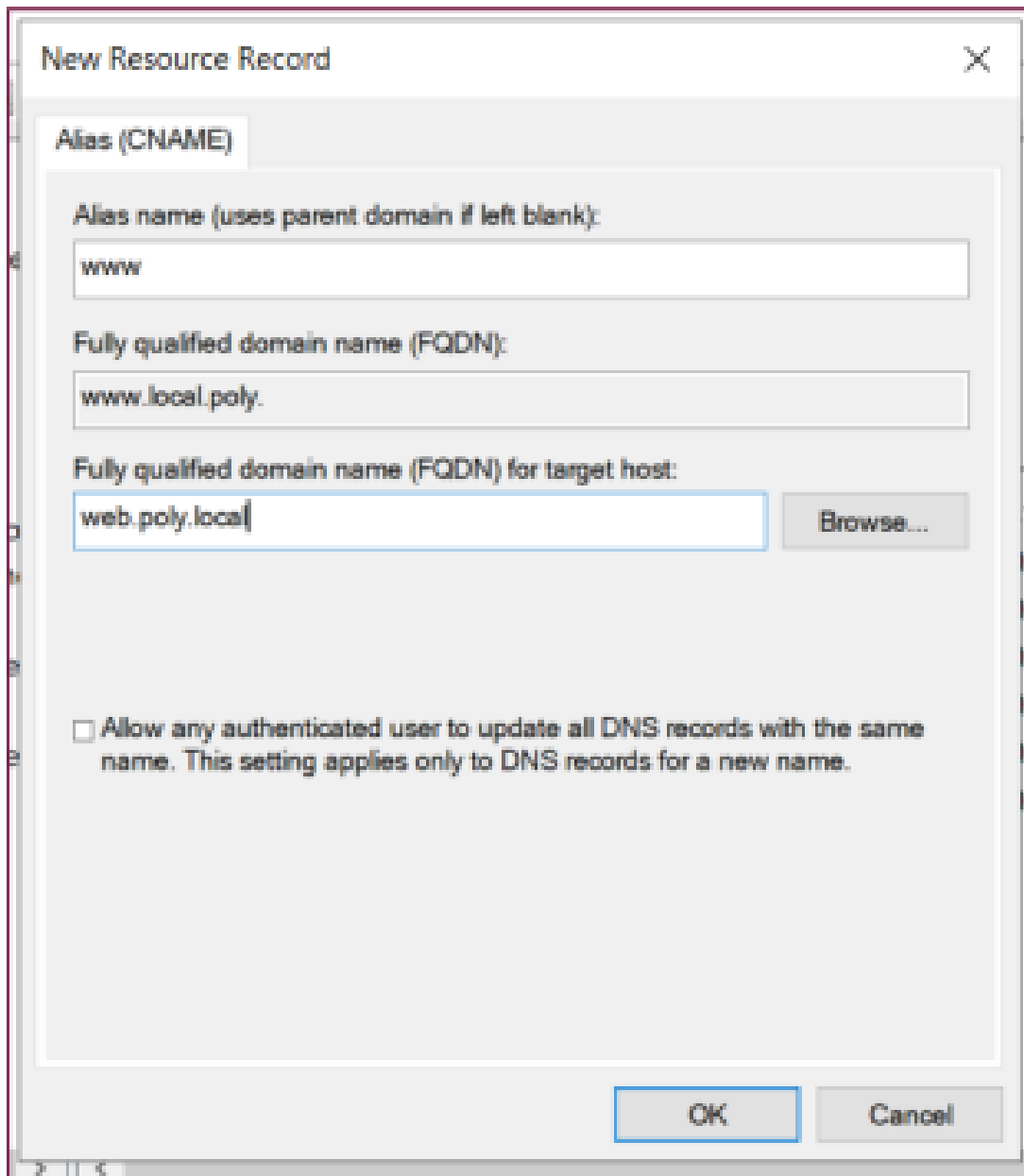


FIGURE 1.7 – CNAME record configuration : alias "www" pointing to target host web.poly.local

- MX Record — mail server mail.poly.local, priority 10, directing mail delivery for the domain
- TXT Record — info = "DNS Lab", demonstrating arbitrary text data storage (commonly used in production for SPF/domain verification records)

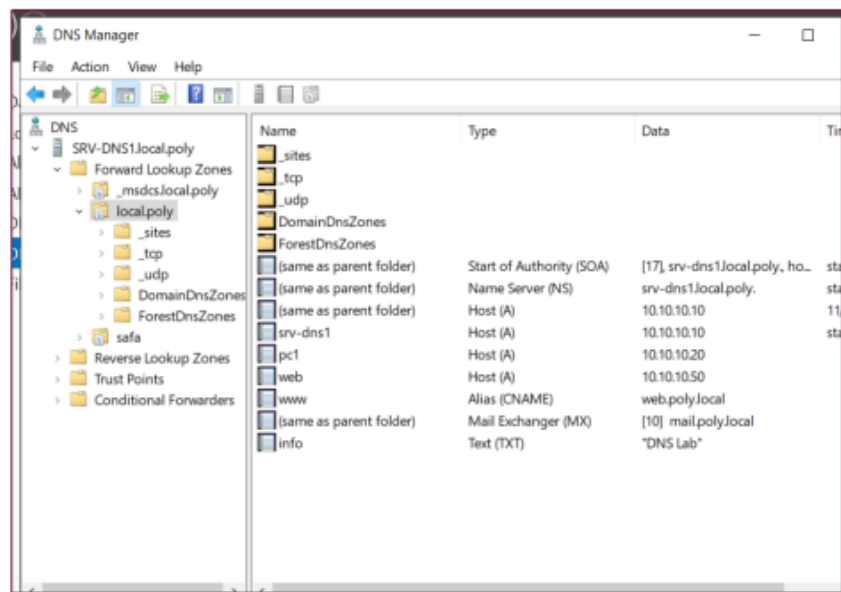


FIGURE 1.8 – DNS Manager overview showing all configured zone records (A, CNAME, MX, TXT)

1.8 7. Secure Dynamic Updates

Both the forward and reverse lookup zones were configured for Secure Only dynamic updates, restricting record updates to authenticated domain-joined machines only. This closes a significant security gap present in "Nonsecure" mode, where DNS entries could otherwise be modified by any client on the network, including untrusted or spoofed sources.

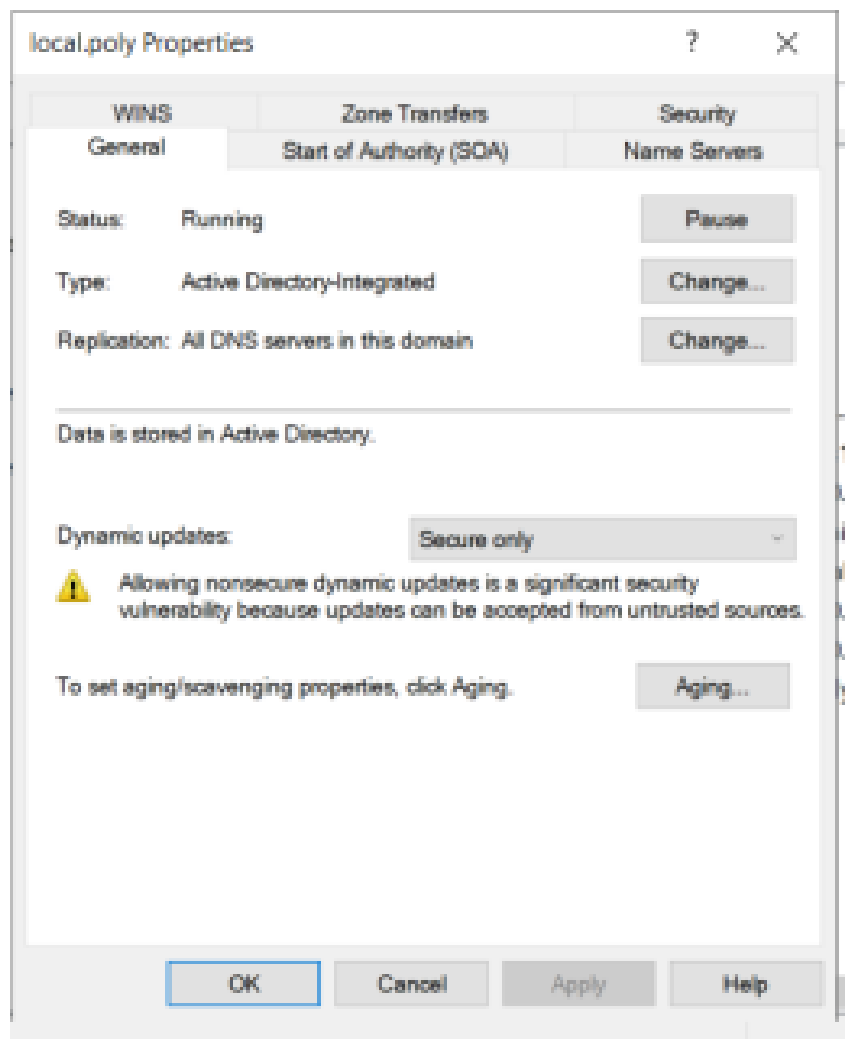


FIGURE 1.9 – Zone properties configured for Secure Only dynamic updates

1.9 8. DNS Scavenging Configuration

Scavenging was configured to automatically remove stale (obsolete) DNS records — a critical maintenance mechanism to prevent zone bloat and outdated entries from accumulating over time, particularly in environments with dynamic registrations.

Configuration steps :

- Enabled globally under Server Properties → Advanced (Enable automatic scavenging of stale records, 7-day scavenging period)
- Aging parameters set via Set Aging/Scavenging for All Zones :
- No-refresh interval : 7 days
- Refresh interval : 7 days

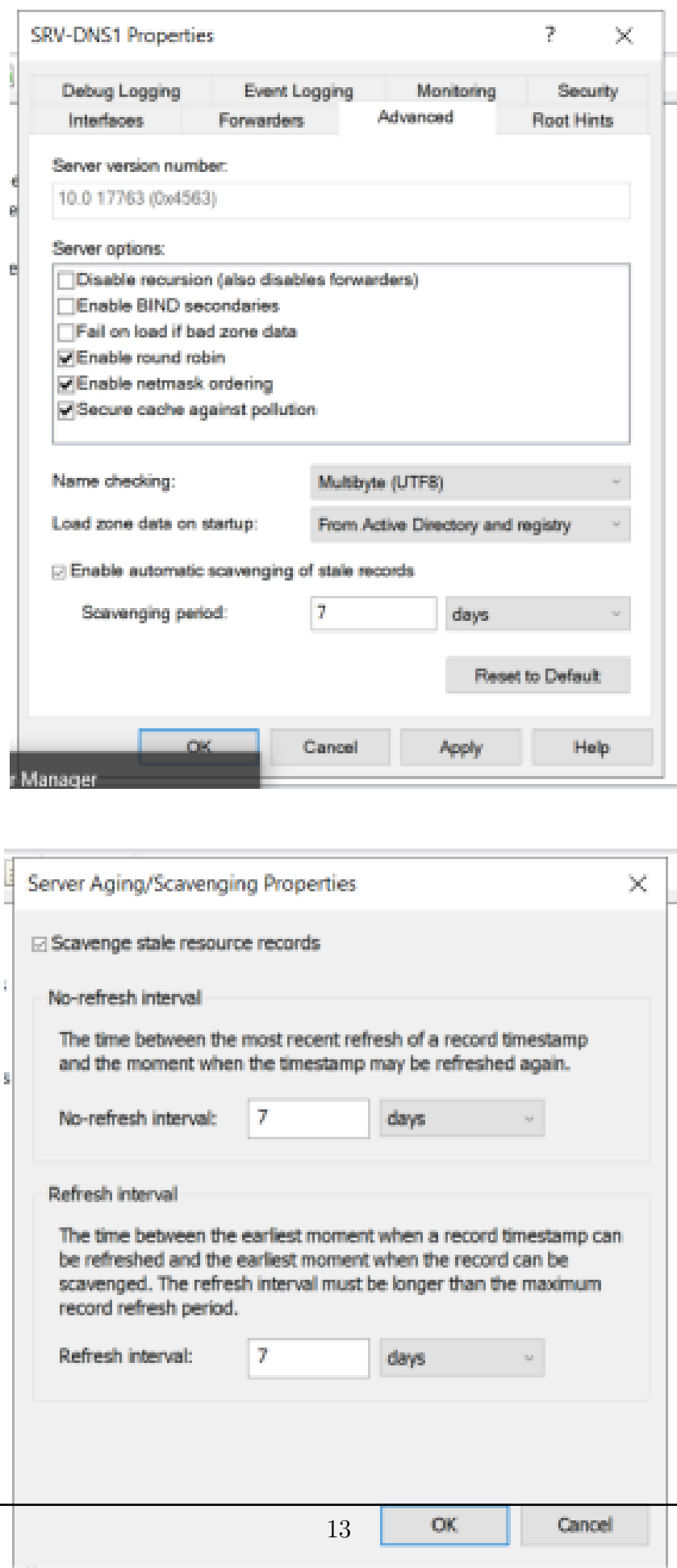


FIGURE 1-10 Enter Caption

1.10 9. Testing & Diagnostics (PowerShell)

All validation in this section was performed via PowerShell, per lab requirements.

1.10.1 9.1 – Forward Resolution (A, CNAME)

Name resolution was tested and validated using multiple complementary tools :

nslookup confirmed correct resolution for pc1.AYA.colcal (→ 10.10.10.20), the reverse query for 10.10.10.20 (→ pc1.aya.colcal), and the CNAME alias www.AYA.colcal (→ web.AYA.colcal, further resolving to 10.10.10.50). Resolve-DnsName, the modern PowerShell cmdlet for DNS queries, confirmed the same results with structured output, including record Type, TTL, and Section — useful for scripting automated DNS health checks rather than parsing raw nslookup text output. Resolve-DnsName -Type A was used to explicitly query for A records, returning both the target host and an NS (Name Server) record, confirming the authoritative server for the zone. Get-Service DNS confirmed the DNS Server service was in a Running state on the server. ipconfig /displaydns was used to inspect the client-side DNS resolver cache, showing cached entries including the pc1.aya.colcal A record and an SRV record for LDAP service discovery (_ldap._tcp...) — confirming the domain controller's service locator records were correctly published and cached.

```

PS C:\Users\Administrator> nslookup pc1.AYA.colcal
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:    pc1.AYA.colcal
Address: 10.10.10.20

PS C:\Users\Administrator> nslookup 10.10.10.20
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:    pc1.aya.colcal
Address: 10.10.10.20

PS C:\Users\Administrator> nslookup WWW.AYA.colcal
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:    web.AYA.colcal
Address: 10.10.10.50
Aliases: WWW.AYA.colcal

PS C:\Users\Administrator> nslookup 10.10.10.50
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:    web.aya.colcal
Address: 10.10.10.50

```

FIGURE 1.11 – nslookup validation of forward and reverse DNS resolution (PC1 and www alias)

```

PS C:\Users\Administrator> Resolve-DnsName pc1.AYA.colcal
Name                                     Type  TTL  Section  IPAddress
----
pc1.AYA.colcal                          A      3600  Answer   10.10.10.20

PS C:\Users\Administrator> Resolve-DnsName www.AYA.colcal
Name                                     Type  TTL  Section  NameHost
----
www.AYA.colcal                          CNAME 3600  Answer   web.AYA.colcal

Name      : web.AYA.colcal
QueryType : A
TTL       : 3600
Section   : Answer
IP4Address : 10.10.10.50

```

FIGURE 1.12 – Resolve-DnsName cmdlet confirming A and CNAME record resolution

```

PS C:\Users\Administrator> Resolve-DnsName -Name web.AYA.colcal -Type A
-----
Name                                     Type  TTL  Section  IPAddress
-----
Test-NetConnection - 10.10.10.50:80
  Attempting TCP connect
  Waiting for response
-----
AYA.colcal                             NS     3600  Answer   win19.AYA.colcal
Name      : win19.AYA.colcal
QueryType : A
TTL       : 3600
Section   : Additional
IP4Address : 10.10.10.10

PS C:\Users\Administrator> ^C
PS C:\Users\Administrator> Get-Service DNS

Status Name      DisplayName
-----
Running DNS      DNS Server

```

FIGURE 1.13 – Resolve-DnsName with NS record lookup and Get-Service confirming DNS service running

```

PS C:\Users\Administrator> ipconfig /displaydns

Windows IP Configuration

pcl.aya.colcal
-----
Record Name . . . . . : pcl.AYA.colcal
Record Type . . . . . : 1
Time To Live . . . . . : 2963
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 10.10.10.20

_ldap._tcp.default-first-site-name._sites.dc._msdcs.aya.colcal
-----
Record Name . . . . . : _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.AYA.colcal
Record Type . . . . . : 33
Time To Live . . . . . : 413
Data Length . . . . . : 16
Section . . . . . : Answer
SRV Record . . . . . : win19.aya.colcal
                        0
                        100
                        389

Record Name . . . . . : win19.aya.colcal
Record Type . . . . . : 1
Time To Live . . . . . : 413
Data Length . . . . . : 4
Section . . . . . : Additional
A (Host) Record . . . : 10.10.10.10

www.aya.colcal
-----
Record Name . . . . . : www.AYA.colcal
Record Type . . . . . : 5

```

FIGURE 1.14 – ipconfig /displaydns showing cached DNS resolver entries including SRV record

1.11 Key Takeaways

This lab provided comprehensive, hands-on experience covering **the full lifecycle of a production-grade DNS deployment** :

- DNS role installation and configuration
- Forward and reverse lookup zone creation and management
- Deployment of essential resource records (A, CNAME, MX, TXT)

- Secure dynamic update configuration to prevent unauthorized record modification
- Scavenging configuration for automated stale-record cleanup
- End-to-end diagnostic validation entirely through PowerShell, reinforcing scripting-based troubleshooting over GUI-only methods

The exercise confirmed the DNS server was fully operational, with both forward and reverse resolution validated from a client machine (PC1) — directly relevant to enterprise network administration, troubleshooting, and security-hardening scenarios (e.g., detecting unauthorized DNS record changes, securing zone transfers).

